

TWO CONTRADICTING LITERATURES, ONE SCALAR

When Does Shift-Invariance Permit Adversarial Robustness?

A threat-matched margin-to-Lipschitz diagnostic η/L

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Thesis: the “invariance hurts robustness” and “invariance helps it” claims are two regimes of one gauge-free, threat-matched ratio η/L ; the field’s invariance metric does not predict robustness, this ratio does, and the “helps” claims are weak-attack artifacts.

The puzzle: same word, opposite verdicts

Does making a CNN more shift-invariant help or hurt adversarial robustness?

“Invariance HURTS”

Ge et al. 2021. A fully shift-invariant classifier can be far *less* robust than a plain one. On a one-pixel problem the achievable margin collapses from a constant to $\mathcal{O}(1/\sqrt{d})$.

Kamath 2021: a provable spatial-vs-adversarial trade-off.

“Invariance HELPS”

Anti-aliasing / TIPS / Wang 2025. Reducing aliasing, learned translation-invariant pooling, and rotation/scale equivariance are *reported to improve* adversarial robustness, argued through an orbit-invariant Lipschitz bound.

“A contradiction in the literature is a flashing sign that some shared assumption is wrong, and the assumption is almost always a word doing too much work. Here the overloaded word is *invariance*.”

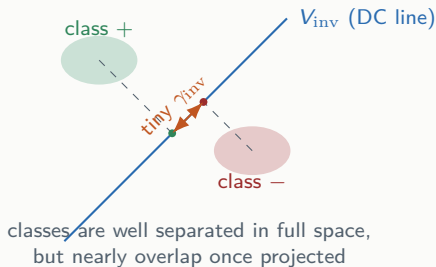
Takeaway. These are not all wrong. They measure different regimes of the *same* underlying quantity. We put them on one axis.

Why invariance can quietly destroy the signal

The linear case is brutally clean

A shift-invariant *linear* classifier must answer the same for a signal and every shift of it. That forces its weights onto the shift group's fixed subspace:

- “The **only** linear shift-invariant feature is the **mean**.” For images: a linear invariant sees nothing but the *average colour*.
- On Ge’s “black vs. white pixel”, it keeps only the DC component; the margin collapses from a constant to $2/\sqrt{d}$ ($\sim 28\times$ at $d=784$).
- **Position lives entirely in the phase**: the power spectrum (magnitude only) is *blind* to location, so even it cannot separate $\pm e_j$.



Takeaway. Invariance keeps only the signal that survives projection onto the invariant subspace; if little survives, robustness must trade off.

The one idea: a margin-to-Lipschitz ratio η/L

Margin that survives the invariance, over its sensitivity to the attack

$$r_2 \geq \frac{\eta}{L} = \frac{\overbrace{\eta}^{\text{margin that survives invariance}}}{\underbrace{L}_{\text{its sensitivity to the attack}}}$$

The certificate. A model is certifiably safe over a ball when it is *decisive* (big margin η) and *gentle* (small Lipschitz L). Robustness attaches to the *ratio*, not to the margin or Lipschitz term alone.

One axis for both camps.

- Large η/L : invariance and robustness coexist.
- Small surviving margin: they trade off.
- Ge / Kamath = the small- η/L corner.

Two properties make η/L the right scalar

Gauge-free and threat-matched

Rescale the logits $f \mapsto c f$: both η and L scale by c , so η/L is unchanged.

The margin and the Lipschitz constant are each *gauge-dependent*, so “robustness improved because the margin grew” is only well-posed once a gauge is fixed. The *ratio* is the honest, gauge-free quantity.

An attack in norm ℓ_p is resisted by sensitivity in the *dual* norm ℓ_q .

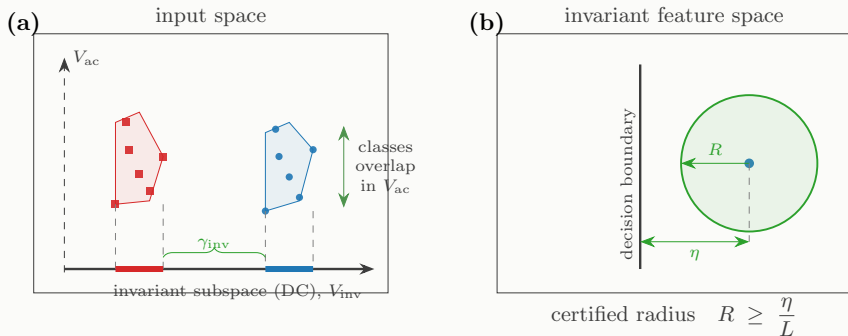
So we always match the Lipschitz term to the attack: $L_q = \mathbb{E} \|\nabla M\|_q$. For an ℓ_∞ attack, use the ℓ_1 gradient norm. Get this wrong and the ratio can even *anti-predict*.

Read it off a trained model on clean inputs: $\eta = \mathbb{E}[\text{logit margin}]$, $L_q = \mathbb{E} \|\nabla(\text{margin})\|_q$.

Takeaway. One organizing principle: invariance changes robustness *only* through η/L , and the attack's norm decides which Lipschitz constant counts.

Structural theory, in one picture

Invariant margin = separation of the data projected onto the invariant subspace



(a) An invariant classifier sees the data only through its projection onto V_{inv} . What separation remains is the invariant margin γ_{inv} ; signal in the orthogonal (AC) subspace is invisible.

(b) A surviving margin η certifies $r_2 \geq \eta/L$. Invariance helps robustness *only* when the discriminative signal survives the projection with positive margin.

Three structural results behind η/L

The invariant linear max-margin *equals* the separation of the DC-projected data, $\gamma_{\text{inv}} = \frac{1}{2} \text{dist}(\text{conv } \Pi_{\text{inv}} X_+, \text{conv } \Pi_{\text{inv}} X_-)$. Recovers Ge's $1/\sqrt{d}$ collapse as a special case; holds for any finite dataset and any finite symmetry group.

A convolution + quadratic activation + global pooling computes the *power spectrum* $\Phi_w(x) = \frac{1}{d} \sum_k |\hat{w}_k|^2 |\hat{x}_k|^2$: a shift-invariant nonlinear feature that can separate data linear invariance cannot. Invariance can help *through the right nonlinear feature*.

Under a reachability condition the one-sided certificate completes to $\eta/L \leq r_2 \leq \eta/\alpha$, with $\kappa = L/\alpha$ (exact, $\kappa=1$, for linear invariants). So η/L determines the robust radius up to a condition number κ .

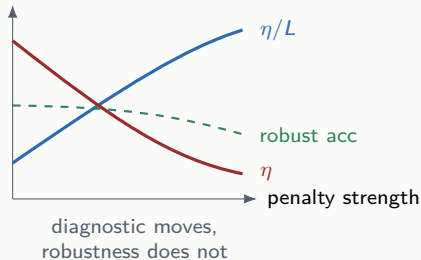
η/L is a diagnostic, *not* a trainable target

The scale-invariance that makes it honest also makes it untrainable

Because η/L is scale-invariant, maximizing it directly is ill-posed: the global optimizer is the **constant classifier** ($\nabla M \equiv 0$), which classifies at chance.

An exact identity (Euler) ties the margin gradient to the margin itself: $\|\nabla M\| \geq M/\|x\|$. Lowering the gradient drags the margin down with it.

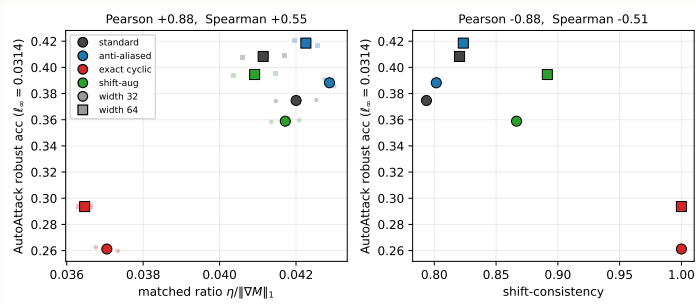
Four first-order attempts to raise η/L (isotropic / anisotropic gradient penalties, ratio maximization, directional penalty) all **collapse the margin** and leave robustness flat or worse.



Takeaway. You can *read* η/L off a model to rank it; you cannot *optimize* it into robustness.

Main empirical result: the dissociation

The field's invariance metric does not predict robustness; η/L does



Left. Threat-matched $\eta/\|\nabla M\|_1$ tracks AutoAttack robust accuracy on adversarially trained CIFAR-10 (Pearson +0.88).

Right. Shift-consistency **anti**-predicts it (−0.88). The exactly invariant model (red) has consistency 1.0 and the *lowest* robust accuracy.

Takeaway. Perfect shift-consistency, worst robustness. Consistency is the *wrong* axis; the threat-matched ratio is the right one.

η/L : threat-matched margin-to-Lipschitz

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The dissociation is not a one-off

Same two laws across datasets, threat models, and a ResNet backbone

condition	training / threat	consist. vs rob.	matched η/L vs rob.
CIFAR-10 dissection	standard, r_2	-0.33	+0.998
CIFAR-10 dissection	PGD-AT, ℓ_∞	-0.88	+0.88
CIFAR-10 dissection	PGD-AT, ℓ_2	-0.81	+0.88
MNIST dissection	PGD-AT, ℓ_∞	-0.44	+0.92
Fashion-MNIST dissection	PGD-AT, ℓ_∞	-0.92	+0.71
ResNet-18 (with TIPS)	PGD-AT, ℓ_∞	-0.79	+0.91
ImageNet-100	Fast-AT, ℓ_∞	-0.87	+0.90
diffusion <i>data</i> axis	PGD-AT, ℓ_∞	near-const	+0.85

Consistency: uninformative to anti-predictive. Threat-matched η/L : predicts throughout.
ImageNet-100: clean accuracy is *flat* across arms, so the deficit is margin geometry, not underfitting.

The strongest result: “invariance helps” is a weak-attack artifact

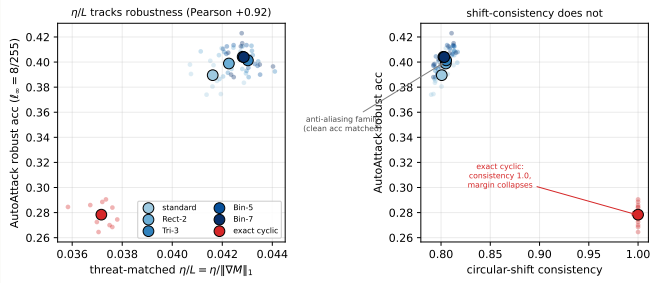
We retrained the contrary papers' own models under their own recipe

arm (standard-trained)	consist.	weak attack		strong attack	
		FGSM ₁	FGSM ₂	AA ₂	AA ₄
standard	0.965	0.477	0.244	0.006	0.000
anti-aliased	0.978	0.472	0.262	0.002	0.000
exact invariant (APS)	1.000	0.497	0.277	0.004	0.000
shift-augmented	0.973	0.532	0.315	0.004	0.000
TIPS + aux losses	0.973	0.493	0.300	0.004	0.000

- Under FGSM the “invariance helps” ordering *does* appear (invariant arms above the baseline).
 - Under AutoAttack every arm collapses to ≤ 0.006 at $\varepsilon = 2/255$ — there is no robustness left to order.
 - Same for Wang 2025's rotation-equivariant net: FGSM 28.5% at their headline budget, but 0.1% under threat-matched AutoAttack.
- ⇒ **The “invariance helps” correlation lives only under weak attacks on undefended models; it vanishes when the attack converges.**

Honest scope: invariance *can* help, when it does not cost margin

A graded anti-aliasing control at matched clean accuracy



Left. Under adversarial training, graded blur-pooling raises robust accuracy at matched clean accuracy, and η/L orders all six arms (+0.92): invariance buys robustness *through the margin*.

Right. Consistency is non-monotone: flat across the anti-aliasing family, then jumps to 1.0 at the exact-cyclic arm, which is the *least* robust.

Takeaway. We do not claim invariance always hurts. η/L , not consistency, tells you *which* invariance helps.

What it means

1. **One axis unifies a split field.** “Invariance hurts” and “invariance helps” are the small- and large- η/L regimes of one gauge-free, threat-matched ratio.
2. **Stop measuring shift-consistency for robustness.** It does not order adversarial robustness and *anti*-predicts it under adversarial training. As an attack-free selection rule it costs ~ 8.8 pts of robust accuracy vs. the oracle; picking by η/L costs 0.06.
3. **Distrust “invariance helps” claims measured under weak attacks.** The effect is an FGSM artifact; it vanishes under AutoAttack and under adversarial training.
4. **η/L is a diagnostic, not a defense.** Read it to rank operators; do not try to optimize it — that collapses the margin.

Honest scope: η/L is a lower-bound certificate (no universal converse), predicts *within* an intervention axis, and the headline is the dissociation, not the near-perfect η/L -radius fit (that part is partly definitional).

How to pitch this

Everything a student needs to give the talk cold

Pitch kit (1/3): the thesis and the 3-beat story

The one-sentence thesis

Two literatures fight over whether shift-invariance helps or hurts adversarial robustness; they are two regimes of one *gauge-free, threat-matched* margin-to-Lipschitz ratio η/L , the field's invariance metric does not predict robustness while η/L does, and the “invariance helps” claims are weak-attack artifacts.

The 3-beat narrative:

- Beat 1. Tension.** Same word, opposite verdicts. Ge: invariance hurts. Anti-aliasing / TIPS / Wang: invariance helps. Nobody put them on one axis.
- Beat 2. Resolution.** One scalar η/L = margin that survives the invariance, over its sensitivity in the attack's dual norm. Large ratio $\Rightarrow$ they coexist; small surviving margin $\Rightarrow$ forced trade-off.
- Beat 3. Payoff.** Consistency *anti*-predicts robustness under adversarial training; η/L predicts it across datasets/threats/backbones; and the “helps” result is an FGSM artifact that dies under AutoAttack.

Pitch kit (2/3): two analogies + the killer result

Two analogies that make it click:

- **“Average colour.”** A linear invariant of an image can see nothing but its *average colour* — everything about *where* things are is thrown away. That is why forcing invariance can collapse the usable margin (Ge’s one-pixel example: margin drops to $2/\sqrt{d}$).
- **“Decisive and gentle.”** A model is certifiably safe when it is *decisive* (big margin) *and gentle* (small Lipschitz). η/L is exactly “decisiveness per unit sensitivity” — and sensitivity must be measured in the attack’s own (dual) norm.

The single result to lead with

Retrain the “invariance helps” models under their own recipe: the exactly invariant arm has *perfect* shift-consistency and looks best under FGSM (0.497 vs. 0.477), yet under AutoAttack every arm collapses to ≤ 0.006 . The advantage was a weak-attack artifact. Under real (adversarial-training) robustness, consistency -0.88 vs. $\eta/L + 0.88$.

Pitch kit (3/3): three questions you *will* get, with answers

Q1. “ $r_2 \geq \eta/L$ is just a loose lower bound — how can it *characterize* robustness?”

A. As a certificate it is one-sided, yes. But (i) empirically it *predicts* robustness (correlations above), and (ii) we prove a two-sided bracket $\eta/L \leq r_2 \leq \eta/\alpha$ that pins r_2 up to a condition number $\kappa = L/\alpha$ (exact, $\kappa=1$, for linear invariants).

Q2. “The dot / DC collapse is a toy. Does it matter on real data?”

A. The same accounting is what the capacity-matched CIFAR / ResNet / ImageNet-100 dissections measure: forcing exact invariance collapses the margin on real data too, and η/L predicts robustness at ImageNet-100 resolution *with clean accuracy held flat*.

Q3. “If η/L predicts robustness, just train on it — new defense?”

A. No. It is scale-invariant, so its maximizer is the constant classifier, and Euler’s identity ties $\|\nabla M\|$ to M . Four first-order attempts all collapse the margin. It is a *diagnostic*, not a target.

ONE LINE TO REMEMBER

Don't ask whether a model is *invariant*.
Ask how much **margin survives the invariance**,
and how **sensitive** that margin is **to the attack**.

$$r_2 \geq \eta/L$$